END TERM EXAMINATION

(CBCS)(SUBJECTIVE TYPE)

B. Tech, 2nd Semester

(AUGUST, 2022)

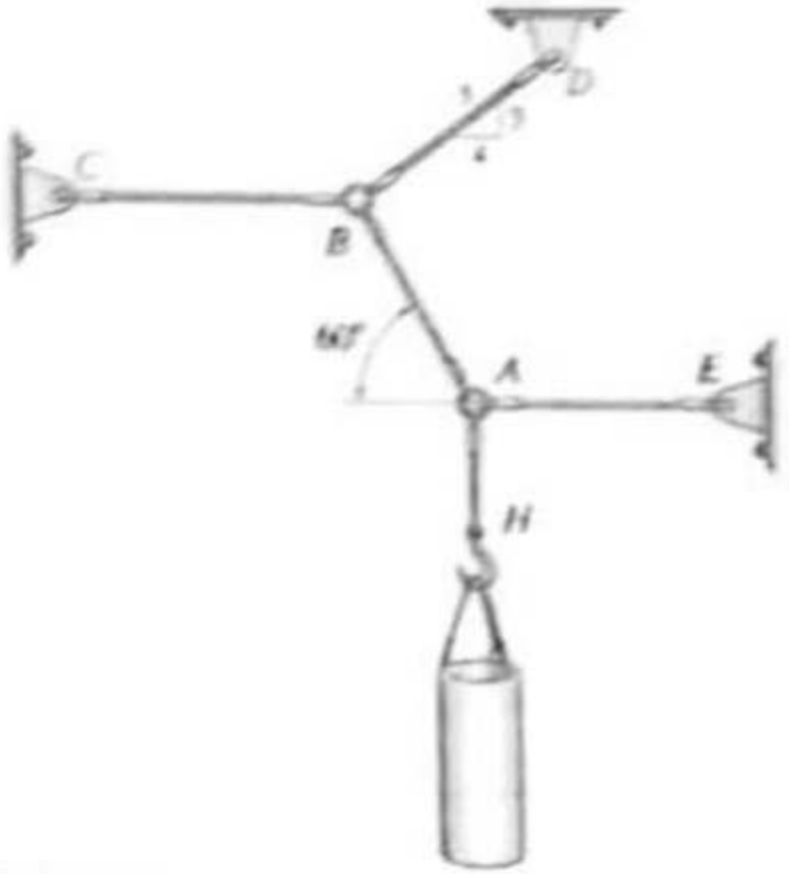
Subject Code: BMA-110	Subject: Engineering Mechanics

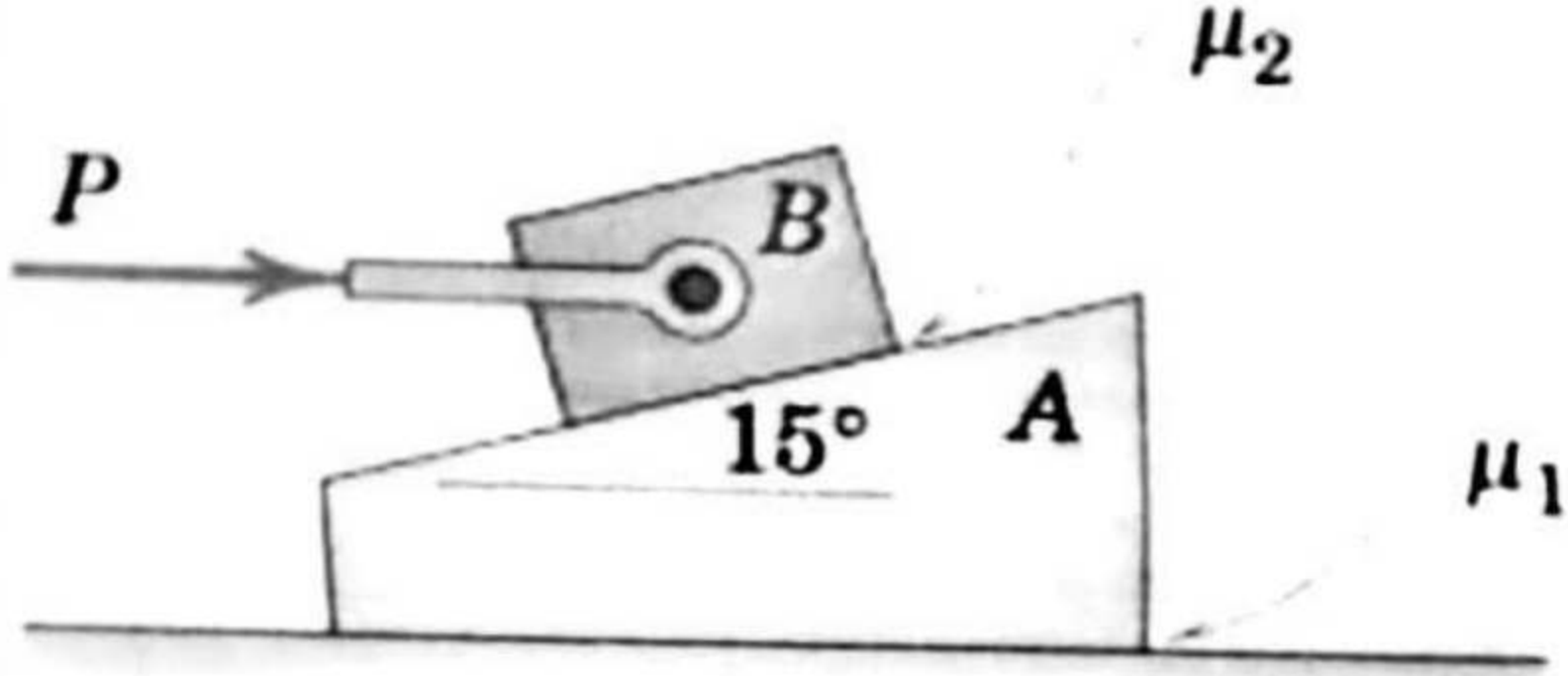
Time :3 Hours

Maximum Marks :60

Note: Q. 1 is compulsory. Attempt one question each from the Units I, II, III & IV.

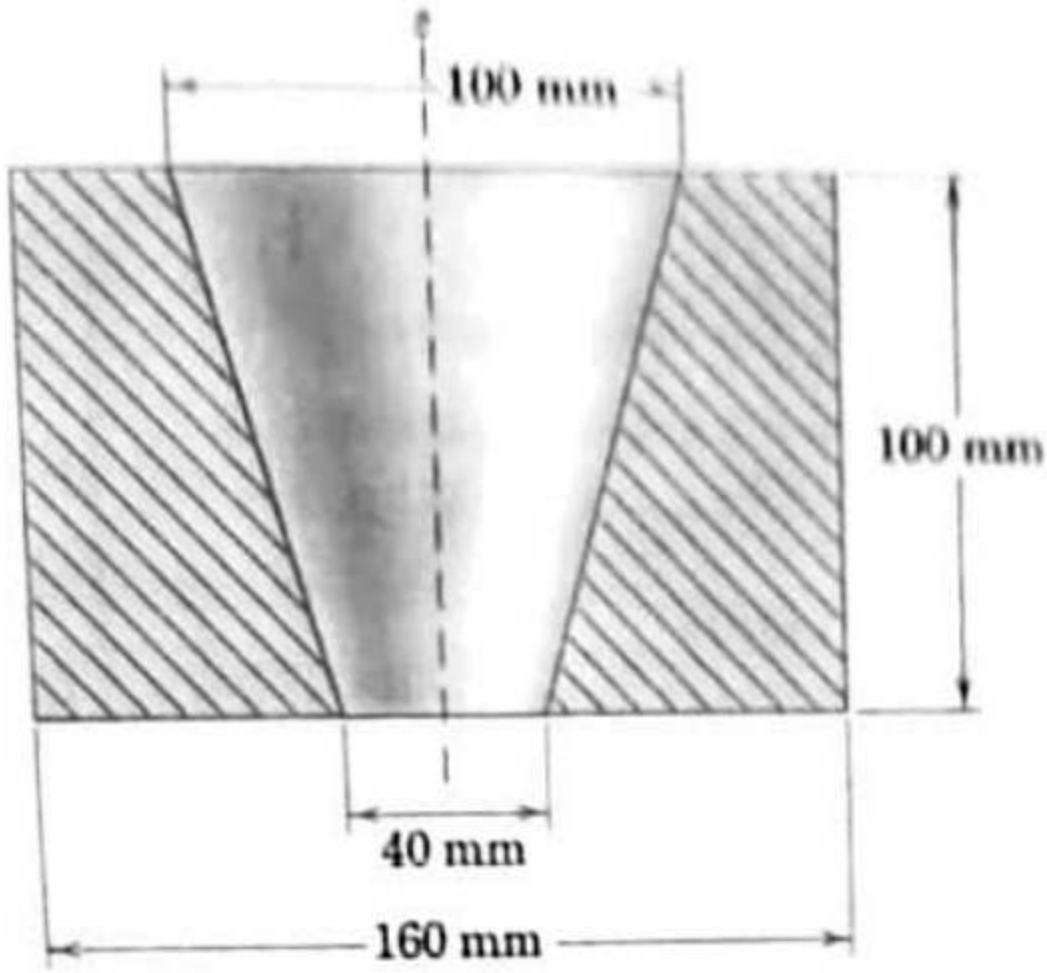
Q1		(20)
a)	State True or False (1 mark each) i) A body is said to be in equilibrium if it has no linear motion. F ii) The like parallel forces are those parallel forces, which are liked by the scientists and engineers. F iii) A couple can be balanced by a force equal to its magnitude. T iv) Lami's Theorem cannot be applied in the case of concurrent forces F v) The motion under gravity is a particular case of motion under constant velocity F vi) The instantaneous centre is a point which is always fixed. F vii) The units of weight are the same as that of force. T viii) A science teacher told to his students that Newton's Third Law of Motion is involved while studying the motion of rockets. Is his statement correct? T ix) Two bodies of masses 10 kg and 15 kg are hung from the ends of an inextensible rope passing over a frictionless pulley. If the masses of both the bodies are doubled the acceleration of the system will also be doubled. F x) When a body moves along a circular path with a uniform velocity, there will be no tangential acceleration. F	
b)	Short Answer Questions (2 marks Each) i. Describe Cone of Friction ii. Explain Lami's theorem. iii. Define the Pappus theorem for Area and Pappus theorem for Volume. iv. Define Rigid Body. v. Explain different types of accelerations involved in curvilinear motion of a particle.	
	UNIT-I	
Q2	Each cord can sustain a maximum tension of 500 N.Determine the largest mass of pipe that can be supported.	(10)

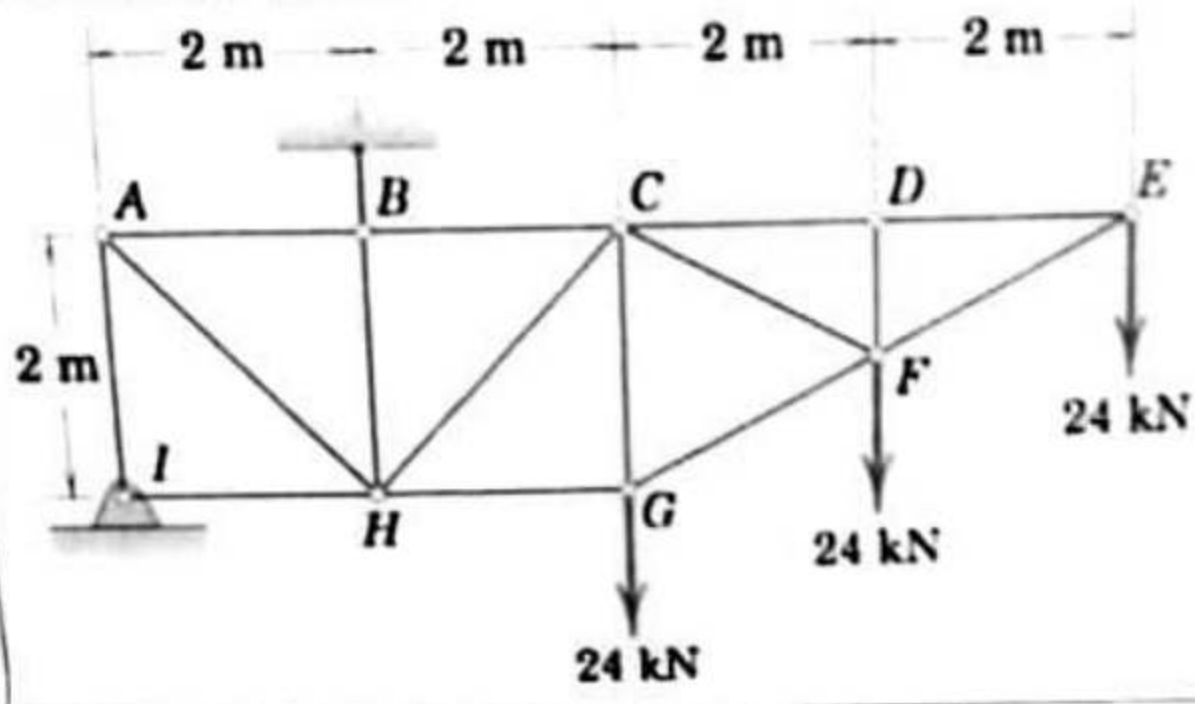


Q3	The horizontal force P = 50 N is applied to the upper block with the system initially stationary. The block masses are m_A = 10 kg and m_B = 5 kg. Determine if and where slippage occurs for the following conditions on the coefficients of static friction: (a) $\mu_1 = 0.40$, $\mu_2 = 0.50$ and (b) $\mu_1 = 0.30$, $\mu_2 = 0.60$. Assume that the coefficients of kinetic friction are 75 percent of the static values.  	(10)

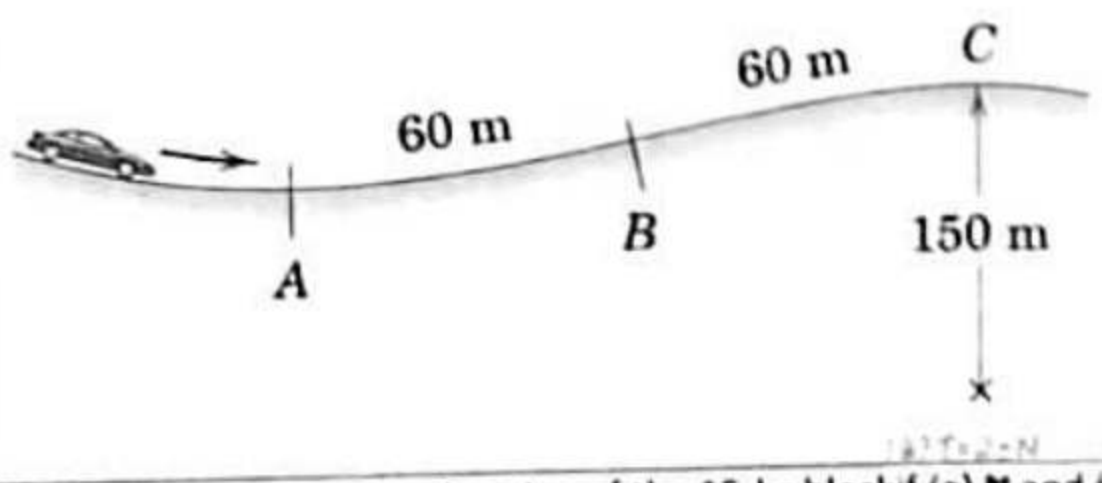
UNIT-II

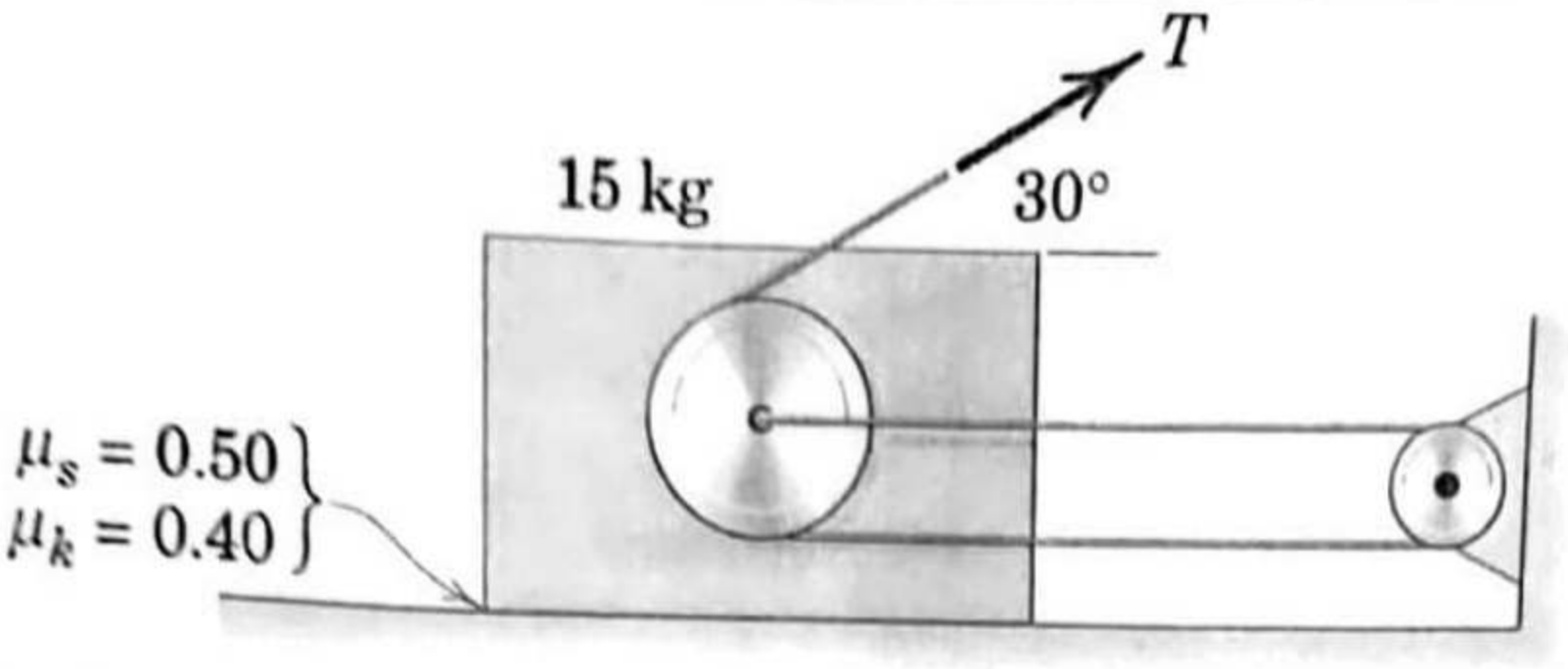
Q4	The cross section shown is for a complete cast-iron body of revolution about the z-axis. Compute its mass m. (Density of cast iron = 7300kg/m3). (The area with the lines is the area that is revolved around z-axis)	(10)



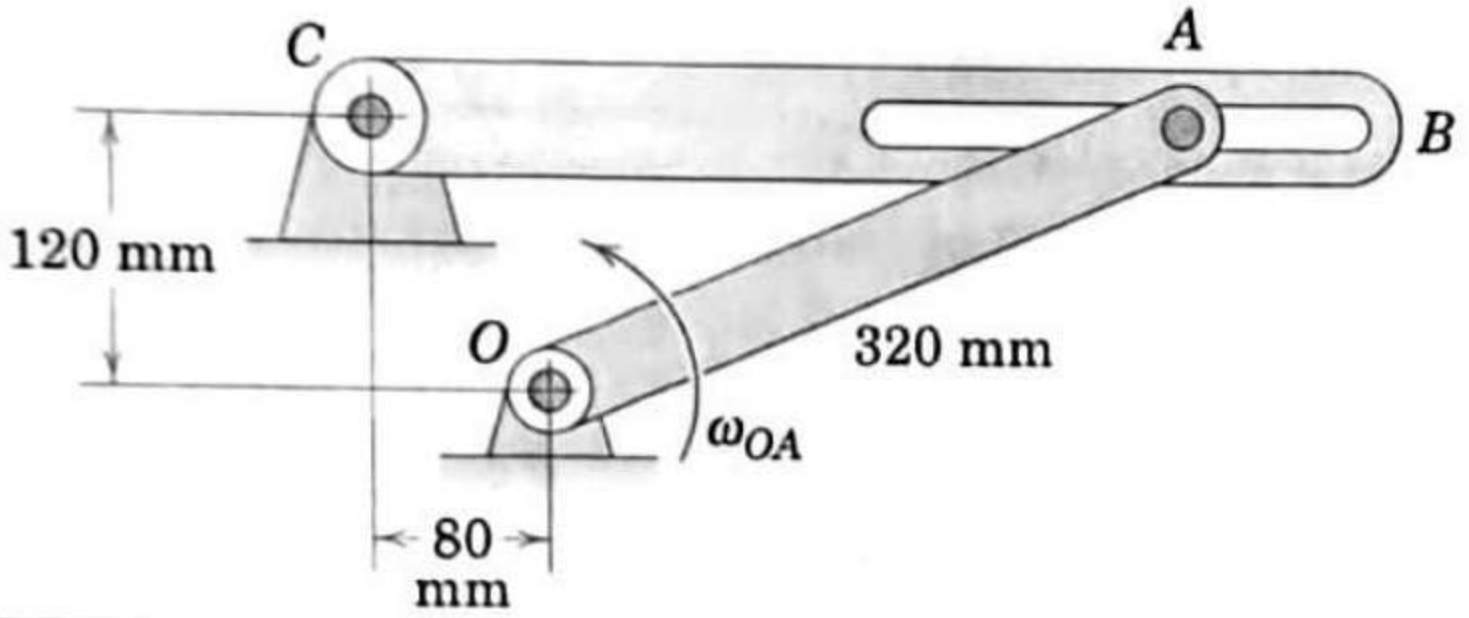
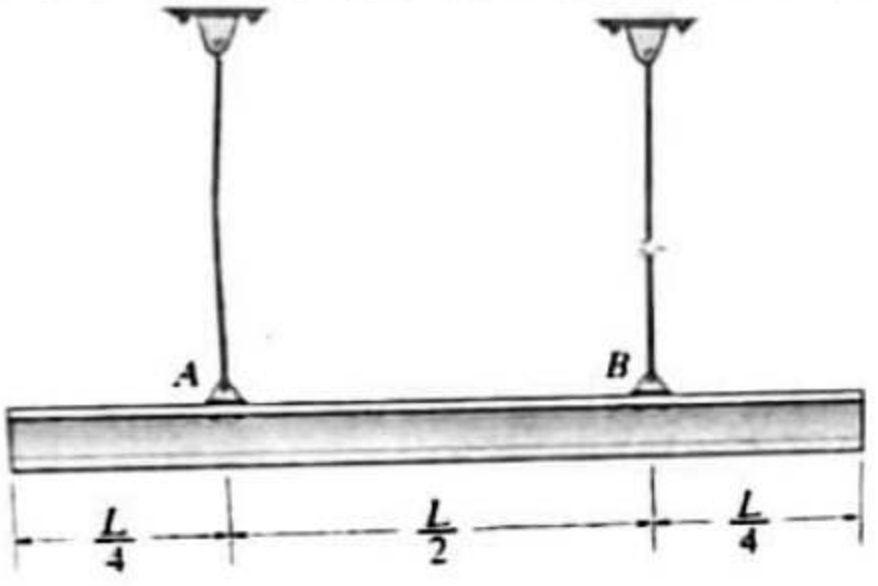
Q5	Determine the forces in members CH and CF. 	(10)

UNIT-III

Q6	To anticipate the dip and hump in the road, the driver of a car applies her brakes to produce a uniform deceleration. Her speed is 100 km/h at the bottom A ofthe dip and 50 km/h at the top C of the hump, which is 120 m along the road from A. If the passengers experience a total acceleration of 3 m/s^2 at A and if the radius ofcurvature of the hump at C is 150 m, calculate (a) the radius of curvature ρ at A, (b)the acceleration at the inflection point B, and (c) the total acceleration at C. 	(10)
Q7	Determine the initial acceleration of the 15-kg blockif (a) 8 and (b) N. The system is initially at rest with no slack in the cable, and themass and friction of the pulleys are negligible.	(10)



UNIT-IV

Q8	Link OA has an angular velocity ω_{OA}=8 rad/s as it passes the position shown. Determine the corresponding angular velocity ω_{CB} of the slotted link CB. Solve by considering the relation between the infinitesimal displacements involved. 	(10)
Q9	The uniform beam has a weight *W*. If it is originallyat rest whilebeing supported at *A* and *B* by cables,determine the tension in cable *A* if cable *B* suddenly fails.Assume the beam is a slender rod. 	(10)